

PSP - Upgrade Goldengate Hub from 12.3 to 19.1

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1. Overview

This document is to provide a step-by-step on how to upgrade Goldengate Hub from 12.3 to 19.1.

2. Make sure IDPS info is update in config file.

- Enter the IDPS endpoints and policy IDs to the GG Hub config file like gg_hub_psp.config

```
["e2e", "prc", "prod"]
ami_us_west_2="ami-08960953abebadc75"
ami_us_east_2="ami-0071f515863fd8541"
vpc_env="prod"
profile="sbg-ppsp-prod"
goldengate_iam="arn:aws:iam::893547637742:instance-profile/goldengate-iam-instance-profile-20190805T115439-InstanceIamProfile-1W9CYYJF6JJQU"
notification_service_arn="arn:aws:sns:us-west-2:893547637742:gg-sbg-ppsp-prod-a-intuit-com"
hosted_zone_id="xx"
domain_name="sbg-ppsp-prod.a.intuit.com"
idps_endpoint="xxxx"
idps_policy_id_us_west_2="xx"
idps_policy_id_us_east_2="xx"
```

3. Detail Steps

Upgrade GG Hub on us-east-2 before us-west-2

3.1 Upgrade GG Hub on us-east-2

- Download the following files https://github.intuit.com/payroll-services-platform/goldengate_hub/tree/master/scripts to your Mac. Please always re-download the files for each db replication setup because they are keeping change for new feature or bug fix.

```
create_gg_hub_ha.sh
gg_hub_psp.config
goldengate-hub-ha.json
```

- Remove auto start flag file on **us-east-2**

```
rm /u01/ogg/scripts/auto_start.flag
```

- Stop extract on **us-west-2**

```
GGSCI (ggpsppf501.sbg-ppsp-ppd.a.intuit.com) 2> stop EPSPSYM1
```

- Wait until no lag on pump on **us-west-2**

```
GGSCI (ggpsppf501.sbg-ppsp-ppd.a.intuit.com) 6> send PPSPSYM2 status
```

Sending STATUS request to EXTRACT PPSPSYM2 ...

EXTRACT PPSPSYM2 (PID 17584)

Current status: Recovery complete: At EOF

Current read position:

Sequence #: 20

RBA: 467640289

Timestamp: 2020-06-15 12:35:10.000000

Extract Trail: /u01/ogg/psppf501/12.3.0.1.4/dirdat/pspsym1/tr

Current write position:

Sequence #: 20

RBA: 467640376

Timestamp: 2020-06-15 12:35:48.265211

Extract Trail: ./dirdat/pspsym1/tr

- Stop pump on **us-west-2**

```
GGSCI (ggpsppf501.sbg-ppsp-ppd.a.intuit.com) 9> stop PPSPSYM2
```

- Wait until all trail files are processed on **us-east-2**

```
GGSCI (ggpspibob1.sbg-ppsp-ppd.a.intuit.com) 4> send RPSPSYM1 status
```

Sending STATUS request to REPLICAT RPSPSYM1 ...

Current status: At EOF

Sequence #: 20

RBA: 467,640,376

0 records in current transaction.

- Stop all processes on **us-east-2**

```
GGSCI (ggpspibob1.sbg-ppsp-ppd.a.intuit.com) 9> stop *
```

- Stop manager on **us-east-2**

```
GGSCI (ggpspibob1.sbg-ppsp-ppd.a.intuit.com) 11> stop mgr
```

- Ensure GG home is correct (19c) and no GG process started yet

```
# rename old 12.3 GG home folder
mv 12.3.0.1.4 12.3.0.1.4_orig

# echo $GG_HOME

# Ensure /l/orcl file is correctly updated with new GG home (19c)
# Ensure tnsnames.ora has right entries

# Ensure no GG process running on host
```

- First check if there is link to file and run the following commands to save the credentials to /u01/tmp temporally on **us-east-2**

```
if there is link exists then
. /l/orcl
mkdir -p /u01/tmp/bfr_19c_upgrade

cp /dev/shm/goldengate/$RDS_SID/dircrd/cwallet.sso /u01/tmp/bfr_19c_upgrade/dircrd_cwallet.sso
cp /dev/shm/goldengate/$RDS_SID/dirwlt/cwallet.sso /u01/tmp/bfr_19c_upgrade/dirwlt_cwallet.sso
cp /dev/shm/goldengate/$RDS_SID/password_file /u01/tmp/bfr_19c_upgrade/

otherwise use below commands
cp $GG_HOME/dircrd/cwallet.sso /u01/tmp/bfr_19c_upgrade/dircrd_cwallet.sso
cp $GG_HOME/dirwlt/cwallet.sso /u01/tmp/bfr_19c_upgrade/dirwlt_cwallet.sso
cp /dev/shm/goldengate/$RDS_SID/password_file /u01/tmp/bfr_19c_upgrade/
```

- Use AWS Console to delete GG Hub Cloudformation stacks starting with goldengate-<RDS>-b and goldengate-<RDS>-b in **us-east-2. Make sure you choose us-east-2 region.**
- Create new GG Hub with Goldengate 19.1 on **us-east-2**

```
chmod +x create_gg_hub_ha.sh
./create_gg_hub_ha.sh
Usage: ./create_gg_hub_ha.sh <AppName> <AppEnv> <DBName> <Region> <VPC> <HA> [<VolumeSize>]
where <AppName> should be psp, payments, etc
      <AppEnv> should be qa, e2e, stg, sbx, prf or prod
      <Region> should be us-west-2, us-east-2, etc
      <HA> should be yes or no
      <VPC> should be vpc-1, vpc-2, etc or Intuit-vpc-1, Intuit-vpc-2, etc
      [<VolumeSize>] is optional. It is EBS volume /u01 size in GB and default is 1000 GB

For non prod example
./create_gg_hub_ha.sh psp prf pspibob1 us-west-2 vpc-2 no

For prod, it would look like
./create_gg_hub_ha.sh psp prod ppsplib1 us-east-2 vpc-1 no
```

- Restore the credentials and upload them to IPDS on GG Hub on **us-east-2**

```

. /l/orcl
APP_ENV=`cat /etc/intu_metadata/app.ini |grep APP_ENV |cut -d=' ' -f2`
ACCOUNT_ID=`cat /etc/intu_metadata/app.ini |grep ACCOUNT_ID |cut -d=' ' -f2`
IDPS_ENDPOINT=`cat /etc/intu_metadata/app.ini |grep IDPS_ENDPOINT |cut -d=' ' -f2`
IDPS_POLICY_ID=`cat /etc/intu_metadata/app.ini |grep IDPS_POLICY_ID|cut -d=' ' -f2`
AWS_REGION=`cat /etc/intu_metadata/app.ini |grep AWS_REGION |cut -d=' ' -f2`

cp /u01/tmp/bfr_19c_upgrade/dircrd_cwallet.sso /dev/shm/goldengate/$RDS_SID/dircrd/cwallet.sso
cp /u01/tmp/bfr_19c_upgrade/dirwlt_cwallet.sso /dev/shm/goldengate/$RDS_SID/dirwlt/cwallet.sso
cp /u01/tmp/bfr_19c_upgrade/password_file /dev/shm/goldengate/$RDS_SID/

stash put-secret --api-key-grant-policy-id ${IDPS_POLICY_ID} --api-endpoint ${IDPS_ENDPOINT} --secret-name
secrets/goldengate/${APP_ENV}/${AWS_REGION}/${RDS_SID/dircrd/cwallet.sso --secret-value /dev/shm/goldengate
/$RDS_SID/dircrd/cwallet.sso

stash put-secret --api-key-grant-policy-id ${IDPS_POLICY_ID} --api-endpoint ${IDPS_ENDPOINT} --secret-name
secrets/goldengate/${APP_ENV}/${AWS_REGION}/${RDS_SID/dirwlt/cwallet.sso --secret-value /dev/shm/goldengate
/$RDS_SID/dirwlt/cwallet.sso

stash put-secret --api-key-grant-policy-id ${IDPS_POLICY_ID} --api-endpoint ${IDPS_ENDPOINT} --secret-name
secrets/goldengate/${APP_ENV}/${AWS_REGION}/db/password_file --secret-value /dev/shm/goldengate/$RDS_SID
/password_file

```

- Copy dirchk directory from 12.1 o 19.1 on **us-east-2**

```

. /l/orcl
cp /u01/ogg/${RDS_SID}/12.3.0.1.4_orig/dirchk/* /u01/ogg/${RDS_SID}/19.1.0.0.1/dirchk/
cp /u01/ogg/${RDS_SID}/12.3.0.1.4_orig/dirprm/* /u01/ogg/${RDS_SID}/19.1.0.0.1/dirprm/
cp /u01/ogg/${RDS_SID}/12.3.0.1.4_orig/diroy/* /u01/ogg/${RDS_SID}/19.1.0.0.1/diroy/

```

- Start manager on **us-east-2**

```
start mgr
```

- Rollover the pump and then start extract and pump on **us-west-2**

```

GGSCI (ggpsppf501.sbg-psp-ppd.a.intuit.com) 14> alter extract PPSPSYM2 etrollover

2020-06-15 13:16:59 INFO    OGG-01520 Rollover performed. For each affected output trail of Version 10 or
higher format, after starting the source extract, issue ALTER EXTSEQNO for that trail's reader (either pump
EXTRACT or REPLICAT) to move the reader's scan to the new trail file; it will not happen automatically.
EXTRACT altered.

GGSCI (ggpsppf501.sbg-psp-ppd.a.intuit.com) 7> start EPSPSYM1

GGSCI (ggpsppf501.sbg-psp-ppd.a.intuit.com) 12> start PPSPSYM2

```

- Recreate Replicat and Alter the replicat to start from the first trail file on **us-east-2**

```
[oracle@ggpspibob1 19.1.0.0.1]$ ls -ltr dirdat/pspsym1/
total 7368
-rw-r----- . 1 oracle oinstall 7542327 Jun 15 13:22 tr000000021

GGSCI (ggpspibob1-a.sbg-ppsp-prod.a.intuit.com as ggt@PSPSPIB1) 8> obey ./diroby/create_ribpp001.oby

GGSCI (ggpspibob1-a.sbg-ppsp-prod.a.intuit.com as ggt@PSPSPIB1) 9> dblogin USERIDALIAS ggtarget

Successfully logged into database.

GGSCI (ggpspibob1-a.sbg-ppsp-prod.a.intuit.com as ggt@PSPSPIB1) 10> add replicat ribpp001, INTEGRATED, EXTTRAIL .
/dirdat/psppib1/tr

REPLICAT (Integrated) added.

GGSCI (ggpspibob1-a.sbg-ppsp-prod.a.intuit.com as ggt@PSPSPIB1) 11> info all
```

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	STOPPED	EIBSP001	00:00:03	01:49:52
EXTRACT	STOPPED	PIBPP001	00:00:00	01:49:52
REPLICAT	STOPPED	RIBPP001	00:00:00	00:00:04

```
GGSCI (ggpspibob1.sbg-ppsp-ppd.a.intuit.com) 1> alter replicat RPSPSYM1 extseqno 21 extrba 0

2020-06-15 13:23:41 INFO OGG-06594 Replicat RPSPSYM1 has been altered. Even the start up position might be
updated, duplicate suppression remains active in next startup. To override duplicate suppression, start
RPSPSYM1 with NOFILTERDUPTRANSACTIONS option.

REPLICAT (Integrated) altered.

GGSCI (ggpspibob1.sbg-ppsp-ppd.a.intuit.com) 3> start RPSPSYM1
Sending START request to MANAGER ...
REPLICAT RPSPSYM1 starting
```

- Setup auto start flag file on **us-east-2**

```
touch /u01/ogg/scripts/auto_start.flag
```

- Restore cronjob on **us-east-2**

```
crontab /u01/ogg/scripts/crontab.txt
```

- Delete the credentials on GG Hub on **us-east-2**

```
rm /u01/tmp/bfr_19c_upgrade/dircred_cwallet.sso
rm /u01/tmp/bfr_19c_upgradedirwlt_cwallet.sso
rm /u01/tmp/bfr_19c_upgrade/password_file
```

3.2 Upgrade GG Hub on us-west-2

- Remove auto start flag file on **us-west-2**

```
rm /u01/ogg/scripts/auto_start.flag
```

- Stop all extracts on us-west-2

```
GGSCI (ggpsppf501.sbg-ppsp-ppd.a.intuit.com) 2> stop e*
```

- Wait until no lag on all pumps on **us-west-2**

```
GGSCI (ggpsppf501.sbg-ppsp-ppd.a.intuit.com) 6> send PPSPSYI2 status
```

Sending STATUS request to EXTRACT PPSPSYI2 ...

EXTRACT PPSPSYI2 (PID 82346)

Current status: Recovery complete: At EOF

Current read position:

Sequence #: 49

RBA: 139220126

Timestamp: 2020-06-17 05:54:33.000000

Extract Trail: /u01/ogg/psppf501/12.3.0.1.4/dirdat/pspsyi1/tr

Current write position:

Sequence #: 49

RBA: 139228285

Timestamp: 2020-06-17 05:55:13.782293

Extract Trail: ./dirdat/pspsyi1/tr

```
GGSCI (ggpsppf501.sbg-ppsp-ppd.a.intuit.com) 7> send PPSPSYM2 status
```

Sending STATUS request to EXTRACT PPSPSYM2 ...

EXTRACT PPSPSYM2 (PID 8670)

Current status: Recovery complete: At EOF

Current read position:

Sequence #: 23

RBA: 364723225

Timestamp: 2020-06-17 05:54:33.000000

Extract Trail: /u01/ogg/psppf501/12.3.0.1.4/dirdat/pspsym1/tr

Current write position:

Sequence #: 23

RBA: 80239326

Timestamp: 2020-06-17 05:55:17.597159

Extract Trail: ./dirdat/pspsym1/tr

- Stop all pumps on us-west-2

```
GGSCI (ggpsppf501.sbg-ppsp-ppd.a.intuit.com) 10> stop p*
```

- Ensure GG home is correct (19c) and no GG process started yet

```
# rename old 12.3 GG home folder
mv 12.3.0.1.4 12.3.0.1.4_orig

# echo $GG_HOME

# Ensure /l/orcl file is correctly updated with new GG home (19c)
# Ensure tnsnames.ora has right entries

# Ensure no GG process running on host
```

- Run the following commands to save the credentials to /u01/tmp temporally on **us-west-2**

```
if there is link exists then
. /l/orcl
mkdir -p /u01/tmp/bfr_19c_upgrade

cp /dev/shm/goldengate/$RDS_SID/dircrd/cwallet.sso /u01/tmp/bfr_19c_upgrade/dircrd_cwallet.sso
cp /dev/shm/goldengate/$RDS_SID/dirwlt/cwallet.sso /u01/tmp/bfr_19c_upgrade/dirwlt_cwallet.sso
cp /dev/shm/goldengate/$RDS_SID/password_file /u01/tmp/bfr_19c_upgrade/

otherwise use below commands
cp $GG_HOME/dircrd/cwallet.sso /u01/tmp/bfr_19c_upgrade/dircrd_cwallet.sso
cp $GG_HOME/dirwlt/cwallet.sso /u01/tmp/bfr_19c_upgrade/dirwlt_cwallet.sso
cp /dev/shm/goldengate/$RDS_SID/password_file /u01/tmp/bfr_19c_upgrade/
```

- Use AWS Console to delete GG Hub Cloudformation stacks starting with goldengate-<RDS>-a and goldengate-<RDS>-b in **us-west-2**. **Make sure you choose us-west-2 region.**

- Create new GG Hub with Goldengate 19.1 on **us-west-2**

```
./create_gg_hub_ha.sh
Usage: ./create_gg_hub_ha.sh <AppName> <AppEnv> <DBName> <Region> <VPC> <HA> [<VolumeSize>]
where <AppName> should be psp, payments, etc
      <AppEnv> should be qa, e2e, stg, sbx, prf or prod
      <Region> should be us-west-2, us-east-2, etc
      <HA> should be yes or no
      <VPC> should be vpc-1, vpc-2, etc or Intuit-vpc-1, Intuit-vpc-2, etc
      [<VolumeSize>] is optional. It is EBS volume /u01 size in GB and default is 1000 GB
```

For prod, the command should like the below,

```
./create_gg_hub_ha.sh psp prod psppp001 us-west-2 vpc-2 no
```

Note: no volume size is needed since it will reuse the original storage

- Restore the credentials and upload them to IPDS on GG Hub on **us-west-2**

```

. /l/orcl
APP_ENV=`cat /etc/intu_metadata/app.ini |grep APP_ENV |cut -d=' ' -f2`
ACCOUNT_ID=`cat /etc/intu_metadata/app.ini |grep ACCOUNT_ID |cut -d=' ' -f2`
IDPS_ENDPOINT=`cat /etc/intu_metadata/app.ini |grep IDPS_ENDPOINT |cut -d=' ' -f2`
IDPS_POLICY_ID=`cat /etc/intu_metadata/app.ini |grep IDPS_POLICY_ID|cut -d=' ' -f2`
AWS_REGION=`cat /etc/intu_metadata/app.ini |grep AWS_REGION |cut -d=' ' -f2`

cp /u01/tmp/bfr_19c_upgrade/dircrd_cwallet.sso /dev/shm/goldengate/$RDS_SID/dircrd/cwallet.sso
cp /u01/tmp/bfr_19c_upgrade/dirwlt_cwallet.sso /dev/shm/goldengate/$RDS_SID/dirwlt/cwallet.sso
cp /u01/tmp/bfr_19c_upgrade/password_file /dev/shm/goldengate/$RDS_SID/

stash put-secret --api-key-grant-policy-id ${IDPS_POLICY_ID} --api-endpoint ${IDPS_ENDPOINT} --secret-name
secrets/goldengate/${APP_ENV}/${AWS_REGION}/${RDS_SID/dircrd/cwallet.sso --secret-value /dev/shm/goldengate
/$RDS_SID/dircrd/cwallet.sso

stash put-secret --api-key-grant-policy-id ${IDPS_POLICY_ID} --api-endpoint ${IDPS_ENDPOINT} --secret-name
secrets/goldengate/${APP_ENV}/${AWS_REGION}/${RDS_SID/dirwlt/cwallet.sso --secret-value /dev/shm/goldengate
/$RDS_SID/dirwlt/cwallet.sso

stash put-secret --api-key-grant-policy-id ${IDPS_POLICY_ID} --api-endpoint ${IDPS_ENDPOINT} --secret-name
secrets/goldengate/${APP_ENV}/${AWS_REGION}/db/password_file --secret-value /dev/shm/goldengate/$RDS_SID
/password_file

```

- Copy dirchk directory from 12.1 o 19.1 on **us-west-2**

```

. /l/orcl

cp /u01/ogg/${RDS_SID}/12.3.0.1.4_orig/dirchk/* /u01/ogg/${RDS_SID}/19.1.0.0.1/dirchk/
cp /u01/ogg/${RDS_SID}/12.3.0.1.4_orig/dirprm/* /u01/ogg/${RDS_SID}/19.1.0.0.1/dirprm/
cp /u01/ogg/${RDS_SID}/12.3.0.1.4_orig/diroby/* /u01/ogg/${RDS_SID}/19.1.0.0.1/diroby/

```

- Check the latest trail file and save the output on **us-east-2**

```

[oracle@ggppspibob1 19.1.0.0.1]$ ls -ltr dirdat/*
dirdat/ppspsym1:
total 1054940
-rw-r-----. 1 oracle oinstall 80239326 Jun 17 05:54 tr000000023

dirdat/ppspsyi1:
total 2715424
-rw-r-----. 1 oracle oinstall 139228285 Jun 17 05:54 tr000000049

```

- Rollover all extracts and pumps on **us-west-2**

```

GGSCI (ggpsppf501.sbg-psp-ppd.a.intuit.com) 2> alter extract EPSPSYI1 etrollover

GGSCI (ggpsppf501.sbg-psp-ppd.a.intuit.com) 4> alter extract EPSPSYM1 etrollover

GGSCI (ggpsppf501.sbg-psp-ppd.a.intuit.com) 5> alter extract PPSPSYI2 etrollover

GGSCI (ggpsppf501.sbg-psp-ppd.a.intuit.com) 6> alter extract PPSPSYM2 etrollover

```

- Recreate and Alter all pumps to start from the last trail file on **us-west-2**

```

# Get extract info
GGSCI (ggpsppf501.sbg-psp-ppd.a.intuit.com) 7> info EPSPSYI1 detail

```



```
EXTRACT    EPSPSYI1  Initialized    2020-06-16 02:20    Status STOPPED
Checkpoint Lag      00:00:01 (updated 01:14:38 ago)
Log Read Checkpoint Oracle Integrated Redo Logs
                  2020-06-17 05:54:33
                  SCN 1952.2121680679 (8385897842471)
```

Target Extract Trails:

Trail Name	Seqno	RBA	Max MB Trail Type
./dirdat/pspsyi1/tr	50	0	500 EXTTRAIL

Drop and create pump

GGSCI (ggpsppib1.sbg-ppsp-prod.a.intuit.com) 9> info all

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	STOPPED			
EXTRACT	STOPPED	EIBPP001	00:00:03	01:11:19
EXTRACT	STOPPED	PIBSP001	00:00:00	01:08:20

GGSCI (ggpsppib1.sbg-ppsp-prod.a.intuit.com) 10> delete extract PIBSP001
Deleted EXTRACT PIBSP001.

GGSCI (ggpsppib1.sbg-ppsp-prod.a.intuit.com) 11> info all

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	STOPPED			
EXTRACT	STOPPED	EIBPP001	00:00:03	01:11:44

GGSCI (ggpsppib1.sbg-ppsp-prod.a.intuit.com) 1> obey ./diroby/create_pibsp001.oby

GGSCI (ggpsppib1.sbg-ppsp-prod.a.intuit.com) 2> add extract pibsp001, EXTTRAILSOURCE ./dirdat/psppib1/tr
EXTRACT added.

GGSCI (ggpsppib1.sbg-ppsp-prod.a.intuit.com) 3> add rmttrail ./dirdat/psppib1/tr, extract pibsp001, MEGABYTES 500

RMTTRAIL added.

GGSCI (ggpsppib1.sbg-ppsp-prod.a.intuit.com) 4>

GGSCI (ggpsppib1.sbg-ppsp-prod.a.intuit.com) 4> info p*

```
EXTRACT    PIBSP001  Initialized    2020-07-23 01:29    Status STOPPED
Checkpoint Lag      00:00:00 (updated 00:00:05 ago)
Log Read Checkpoint File ./dirdat/psppib1/tr0000000000
                  First Record RBA 0
```

Alter pump to start from same detail got from "info extract" command output

GGSCI (ggpsppib1.sbg-ppsp-prod.a.intuit.com) 5> alter extract PIBSP001 extseqno 5443 extrba 0
EXTRACT altered.

GGSCI (ggpsppf501.sbg-ppsp-ppd.a.intuit.com) 9> alter extract PPSPSYI2 extseqno 50 extrba 0
EXTRACT altered.

GGSCI (ggpsppf501.sbg-ppsp-ppd.a.intuit.com) 10> info PPSPSYI2

```
EXTRACT    PPSPSYI2  Initialized    2020-06-17 07:09    Status STOPPED
Checkpoint Lag      00:00:00 (updated 00:00:03 ago)
```

```
Log Read Checkpoint File /u01/ogg/psppf501/12.3.0.1.4/dirdat/pssym1/tr000000050
First Record RBA 0
```

```
GGSCI (ggpsppf501.sbg-ppp-a.intuit.com) 12> info EPSPSYM1 detail
```

```
EXTRACT EPSPSYM1 Initialized 2020-06-15 23:50 Status STOPPED
Checkpoint Lag 00:00:02 (updated 01:15:40 ago)
Log Read Checkpoint Oracle Integrated Redo Logs
2020-06-17 05:54:33
SCN 1950.285256344 (8375471483544)
```

Target Extract Trails:

Trail Name	Seqno	RBA	Max MB Trail Type
./dirdat/pssym1/tr	24	0	500 EXTTRAIL

```
GGSCI (ggpsppf501.sbg-ppp-a.intuit.com) 13> alter extract PPSPSYM2 extseqno 24 extrba 0
EXTRACT altered.
```

```
GGSCI (ggpsppf501.sbg-ppp-a.intuit.com) 14> info PPSPSYM2
```

```
EXTRACT PPSPSYM2 Initialized 2020-06-17 07:10 Status STOPPED
Checkpoint Lag 00:00:00 (updated 00:00:05 ago)
Log Read Checkpoint File /u01/ogg/psppf501/12.3.0.1.4/dirdat/pssym1/tr000000024
First Record RBA 0
```

- Start manager, extracts and pumps on **us-west-2**

```
GGSCI (ggpsppf501.sbg-ppp-a.intuit.com) 16> start mgr
Manager started.
```

```
GGSCI (ggpsppf501.sbg-ppp-a.intuit.com) 17> start e*
```

```
Sending START request to MANAGER ...
EXTRACT EPSPSYI1 starting
```

```
Sending START request to MANAGER ...
EXTRACT EPSPSYM1 starting
```

```
GGSCI (ggpsppf501.sbg-ppp-a.intuit.com) 18> start p*
```

```
Sending START request to MANAGER ...
EXTRACT PPSPSYI2 starting
```

```
Sending START request to MANAGER ...
EXTRACT PPSPSYM2 starting
```

- Alter the replicat to start the oldest trail file generated from 19.1 GG on **us-east-2**

```
# Make sure that replicat finishes apply on the 12.1 trail files (see EOF)
GGSCI (ggpspiob1.sbg-psp-ppd.a.intuit.com) 4> send RPSPSYM1 status
Sending STATUS request to REPLICAT RPSPSYM1 ...
  Current status: At EOF
  Sequence #: 20
  RBA: 467,640,376
  0 records in current transaction.

[oracle@ggpspiob1 19.1.0.0.1]$ ls -ltr dirdat/pspsyi1/|tail
total 2759044
-rw-r-----. 1 oracle oinstall 139228285 Jun 17 07:25 tr0000000049
-rw-r-----. 1 oracle oinstall      1272 Jun 17 07:25 tr0000000050
-rw-r-----. 1 oracle oinstall  44660052 Jun 17 07:25 tr0000000051
[oracle@ggpspiob1 19.1.0.0.1]$ ls -ltr dirdat/pspsym1/|tail
total 1076976
-rw-r-----. 1 oracle oinstall  80239326 Jun 17 07:25 tr0000000023
-rw-r-----. 1 oracle oinstall      1272 Jun 17 07:25 tr0000000024
-rw-r-----. 1 oracle oinstall  22558522 Jun 17 07:26 tr0000000025

GGSCI (ggpspiob1.sbg-psp-ppd.a.intuit.com) 2> stop RPSPSYM1

GGSCI (ggqbosc04-a.sbg-qbo-ppd.a.intuit.com) 8> info all

Program      Status      Group      Lag at Chkpt  Time Since Chkpt
MANAGER      RUNNING
REPLICAT     STOPPED     RPQBPC04    00:00:00      00:00:01

GGSCI (ggpspiob1.sbg-psp-ppd.a.intuit.com) 4> alter replicat RPSPSYI1 extseqno 51 extrba 0

2020-06-17 07:32:12 INFO      OGG-06594 Replicat RPSPSYI1 has been altered. Even the start up position might be
updated, duplicate suppression remains active in next startup. To override duplicate suppression, start
RPSPSYI1 with NOFILTERDUPTRANSACTIONS option.

REPLICAT (Integrated) altered.

GGSCI (ggpspiob1.sbg-psp-ppd.a.intuit.com) 5> alter replicat RPSPSYM1 extseqno 25 extrba 0

2020-06-17 07:32:38 INFO      OGG-06594 Replicat RPSPSYM1 has been altered. Even the start up position might be
updated, duplicate suppression remains active in next startup. To override duplicate suppression, start
RPSPSYM1 with NOFILTERDUPTRANSACTIONS option.

REPLICAT (Integrated) altered.

GGSCI (ggpspiob1.sbg-psp-ppd.a.intuit.com) 7> start r*

Sending START request to MANAGER ...
REPLICAT RPSPSYI1 starting

Sending START request to MANAGER ...
REPLICAT RPSPSYM1 starting
```

- Drop and recreate extract and pump on **us-east-2**

```
GGSCI (ggpspibob1.sbg-ppsp-ppd.a.intuit.com) 1> info all

GGSCI (ggpspibob1.sbg-ppsp-ppd.a.intuit.com) 2> delete EPSPSYM2

GGSCI (ggpspibob1.sbg-ppsp-ppd.a.intuit.com) 2> delete PPSPSYM1

GGSCI (ggpspibob1.sbg-ppsp-ppd.a.intuit.com as ggs@PQB0SC04) 25> unregister extract EPSPSYM2 database

GGSCI (ggpspibob1.sbg-ppsp-ppd.a.intuit.com) 3> obey diroby/create_epspsym2.oby

GGSCI (ggpspibob1.sbg-ppsp-ppd.a.intuit.com) 4> obey diroby/create_ppspsym1.oby

GGSCI (ggpspibob1.sbg-ppsp-ppd.a.intuit.com as ggs@PQB0SC04) 26> register extract EPSPSYM2 database

GGSCI (ggpspibob1.sbg-ppsp-ppd.a.intuit.com as ggs@PQB0SC04) 27> start e*

GGSCI (ggpspibob1.sbg-ppsp-ppd.a.intuit.com as ggs@PQB0SC04) 28> start p*
```

- Confirm the trail file is shipped from us-east-2 on **us-west-2**

```
[oracle@ggppsppf501 19.1.0.0.1]$ ls -ltr dirdat/pspsym2
```

- Setup auto start flag file on **us-west-2**

```
touch /u01/ogg/scripts/auto_start.flag
```

- Restore cronjob on **us-west-2**

```
crontab /u01/ogg/scripts/crontab.txt
```

- Delete the credentials on GG Hub on **us-west-2**

```
rm /u01/tmp/bfr_19c_upgrade/dircred_cwallet.sso
rm /u01/tmp/bfr_19c_upgrade/dirwlt_cwallet.sso
rm /u01/tmp/bfr_19c_upgrade/password_file
```

- Clean up the old GG directory on both **us-west-2** and **us-east-2** **NEXT DAY**

```
. /l/orcl
cd /u01/ogg/${RDS_SID}
rm -r 12.3.0.1.4_orig
```

4. Appendix

4.1 Switch the GG Hub creation code between Goldengate 12.1 and 19.1

Under the normal circumstance, you don't need to switch the GG Hub creation code between Goldengate 12.1 and 19.1. You can run the following commands to do so if needed.

- Switch from 12.1 to 19.1

```
change code in create_gg_hub_ha.sh accordingly as well as below files.
create_gg_hub_ha.sh
gg_hub_psp.config
goldengate-hub-ha.json

chmod +x create_gg_hub_ha.sh
```

- Switch from 19.1 to 12.1.

```
change code in create_gg_hub_ha.sh accordingly as well as below files.
create_gg_hub_ha.sh
gg_hub_psp.config
goldengate-hub-ha.json

chmod +x create_gg_hub_ha.sh
```

4.2 Downgrade GG Hub on us-west-2

You should downgrade GG Hub on us-west-2 before us-east-2

- Run the following SQL command on the primary RDS on **us-west-2**

```
SQL> execute DBMS_LOGMNR_D.BUILD( options => DBMS_LOGMNR_D.STORE_IN_REDO_LOGS);

PL/SQL procedure successfully completed.

SQL> SELECT first_change#, to_char(first_time, 'yyyy-mm-dd hh24:mi:ss') FROM v$sarchived_log WHERE
dictionary_begin = 'YES' AND standby_dest = 'NO' AND name IS NOT NULL AND status = 'A';

FIRST_CHANGE# TO_CHAR(FIRST_TIME,
-----
5331439680 2019-04-22 13:33:25
...
5374736675 2019-05-08 09:42:44
5377816740 2019-05-09 10:01:50

22 rows selected.
```

Note: we use the last row's FIRST_CHANGE# (5377816740 in the example above) later on

- Cleanup dirchk and dirdat under GG 12.3 on **us-west-2**

```
. /l/orcl
[oracle@ggpbop04-a 19.1.0.0.1]$ rm -r /u01/ogg/${RDS_SID}/12.3.0.1.4/dirchk/*
[oracle@ggpbop04-a 19.1.0.0.1]$ rm -r /u01/ogg/${RDS_SID}/12.3.0.1.4/dirdat/*
```

- Stop all extracts on **us-west-2**

```
GGSCI (ggpbop04-a.sbg-qbo-ppd.a.intuit.com) 2> stop e*
```

- Stop all pumps after no lag on **us-west-2**

```
GGSCI (ggpqbopc04-a.sbg-qbo-ppd.a.intuit.com) 5> send PPQBSC04 status

Sending STATUS request to EXTRACT PPQBSC04 ...
EXTRACT PPQBSC04 (PID 11189)
  Current status: Recovery complete: At EOF

GGSCI (ggpqbopc04-a.sbg-qbo-ppd.a.intuit.com) 4> send PPIWPC04 status

Sending STATUS request to EXTRACT PPIWPC04 ...
EXTRACT PPIWPC04 (PID 25375)
  Current status: Recovery complete: At EOF

GGSCI (ggpqbopc04-a.sbg-qbo-ppd.a.intuit.com) 4> stop p*
```

- Check the extracts SCN on **us-west-2**

```
GGSCI (ggpqbopc04-a.sbg-qbo-ppd.a.intuit.com) 4> info EPQBPC04

EXTRACT   EPQBPC04   Last Started 2019-05-09 09:52   Status STOPPED
Checkpoint Lag      00:00:05 (updated 00:03:02 ago)
Log Read Checkpoint Oracle Integrated Redo Logs
                   2019-05-09 10:02:54
                   SCN 1.1082850947 (5377818243)

GGSCI (ggpqbopc04-a.sbg-qbo-ppd.a.intuit.com) 7> info EPIAPC04

EXTRACT   EPIAPC04   Last Started 2019-08-27 21:41   Status STOPPED
Checkpoint Lag      00:00:02 (updated 00:01:11 ago)
Log Read Checkpoint Oracle Integrated Redo Logs
                   2019-09-05 10:23:20
                   SCN 1.1346806789 (5641774085)
```

Note: We will use the SCNs (5377818243 for EPQBPC04 and 5641774085 for EPIAPC04) later on

- Use the command in 4.1 above to switch from 19.1 to 12.3
- Use AWS Console to delete GG Hub Cloudformation stacks starting with goldengate-<RDS>-b and goldengate-<RDS>-b in **us-west-2**. Make sure you choose **us-west-2** region.
- Create new GG Hub with Goldengate 12.3 on **us-west-2**

```
chmod +x create_gg_hub_ha.sh
./create_gg_hub_ha.sh
Usage: ./create_gg_hub_ha.sh <AppName> <AppEnv> <DBName> <Region> <VPC> <HA> [<VolumeSize>]
      where <AppName> should be qbo, payments, etc
            <AppEnv> should be qa, e2e, stg, sbx, prf or prod
            <Region> should be us-west-2, us-east-2, etc
            <HA> should be yes or no
            <VPC> should be vpc-1, vpc-2, etc or Intuit-vpc-1, Intuit-vpc-2, etc
            [<VolumeSize>] is optional. It is EBS volume /u01 size in GB and default is 1000 GB

For non prod example
./create_gg_hub_ha.sh psp prf pspibob1 us-west-2 vpc-2 no

For prod, it would look like
./create_gg_hub_ha.sh psp prod pspspib1 us-east-2 vpc-1 no

Note: no volume size is needed since it will reuse the original storage
```

- Confirm the replicat has finished processing all old trail files and the status shows at EOF and take a note on Integrated Replicat low watermark on us-east-2

```
GGSCI (ggpqbosc04-a.sbg-qbo-ppd.a.intuit.com) 3> send RPQBPC04 status

Sending STATUS request to REPLICAT RPQBPC04 ...
Current status: At EOF
Sequence #: 27
RBA: 122,685
0 records in current transaction.

GGSCI (ggpqbosc04-a.sbg-qbo-ppd.a.intuit.com) 5> obey diroby/dblogin_t.oby

GGSCI (ggpqbosc04-a.sbg-qbo-ppd.a.intuit.com) 6> dblogin USERIDALIAS ggtarget

Successfully logged into database.

GGSCI (ggpqbosc04-a.sbg-qbo-ppd.a.intuit.com as ggt@PQBOSC04) 7> info replicat RPQBPC04 detail

REPLICAT RPQBPC04 Last Started 2019-05-09 09:57 Status RUNNING
INTEGRATED
Checkpoint Lag 00:00:00 (updated 00:00:06 ago)
Process ID 26398
Log Read Checkpoint File /u01/ogg/pqbosc04/19.1.0.0.1/dirdat/pqbopc04/tr000027
2019-05-09 10:02:51.045581 RBA 122685

INTEGRATED Replicat
DBLOGIN Provided, inbound server name is OGG$RPQBPC04 in ATTACHED state

Current Log BSN value: 5374729495

Low Watermark CSN value: 5377818197
(All source transactions prior to this scn have been applied)

High Watermark CSN value: 5377818221
(Some source transactions between this scn and the low watermark may have been applied)
```

Note: We will the the Low Watermark CSN (5377818197 in the example above) later on

- Stop the repliact, rename trail file location directory and create new the directory on us-east-2

```
GGSCI (ggpqbosc04-a.sbg-qbo-ppd.a.intuit.com) 1> info all

Program      Status      Group      Lag at Chkpt  Time Since Chkpt
MANAGER      RUNNING
REPLICAT     RUNNING     RPQBPC04   00:00:00      00:00:02

GGSCI (ggpqbosc04-a.sbg-qbo-ppd.a.intuit.com) 2> stop r*

Sending STOP request to REPLICAT RPQBPC04 ...
Request processed.

[oracle@ggpqbosc04-a dirdat]$ mv pqbopc04 pqbopc04_tmp
[oracle@ggpqbosc04-a dirdat]$ mkdir pqbopc04
```

- Create temp extract obey files on us-west-2

```
[oracle@ggpbop04-a diroby]$ cp create_epqbpc04.oby create_epqbpc04_tmp.oby
[oracle@ggpbop04-a diroby]$ vi create_epqbpc04_tmp.oby

[oracle@ggpbop04-a diroby]$ diff create_epqbpc04.oby create_epqbpc04_tmp.oby
2c2
< add extract epqbpc04, INTEGRATED TRANLOG, BEGIN NOW
---
> add extract epqbpc04, INTEGRATED TRANLOG, SCN 5377818243
```

Note: 5377818243 is the extract's Log Read Checkpoint

```
[oracle@ggpbop04-a diroby]$ cp create_epiapc04.oby create_epiapc04_tmp.oby
[oracle@ggpbop04-a diroby]$ vi create_epiapc04_tmp.oby

[oracle@ggpbop04-a diroby]$ diff create_epiapc04.oby create_epiapc04_tmp.oby
2c2
< add extract epiapc04, INTEGRATED TRANLOG, BEGIN NOW
---
> add extract epiapc04, INTEGRATED TRANLOG, SCN 5641774085
```

Note: 5641774085 is the extract's Log Read Checkpoint

- Recreate all extracts on us-west-2

```
GGSCI (ggpbop04-a.sbg-qbo-ppd.a.intuit.com) 1> obey diroby/dblogin_s.oby
GGSCI (ggpbop04-a.sbg-qbo-ppd.a.intuit.com) 2> dblogin USERIDALIAS ggs
Successfully logged into database.

GGSCI (ggpbop04-a.sbg-qbo-ppd.a.intuit.com as ggs@PQBOPC04) 3> info all

Program      Status      Group      Lag at Chkpt  Time Since Chkpt
MANAGER      STOPPED

GGSCI (ggpbop04-a.sbg-qbo-ppd.a.intuit.com as ggs@PQBOPC04) 4> unregister EXTRACT EPQBPC04 database
ERROR: EXTRACT EPQBPC04 must be registered with the database to perform this operation.

GGSCI (ggpbop04-a.sbg-qbo-ppd.a.intuit.com as ggs@PQBOPC04) 5> obey diroby/create_epqbpc04_tmp.oby
GGSCI (ggpbop04-a.sbg-qbo-ppd.a.intuit.com as ggs@PQBOPC04) 6> dblogin USERIDALIAS ggs
Successfully logged into database.

GGSCI (ggpbop04-a.sbg-qbo-ppd.a.intuit.com as ggs@PQBOPC04) 7> add extract epqbpc04, INTEGRATED TRANLOG, SCN
5377818243

EXTRACT added.

GGSCI (ggpbop04-a.sbg-qbo-ppd.a.intuit.com as ggs@PQBOPC04) 8> add exttrail ./dirdat/pqbop04/tr extract
epqbpc04, MEGABYTES 500

EXTTRAIL added.

GGSCI (ggpbop04-a.sbg-qbo-ppd.a.intuit.com as ggs@PQBOPC04) 9> register EXTRACT EPQBPC04 database scn
5377816740
Extract EPQBPC04 successfully registered with database at SCN 5377816740.

Where 5377816740 is the last change# from the query
```


GGSCI (ggpqbopc04-a.sbg-qbo-ppd.a.intuit.com as ggs@PQB0PC04) 12> unregister EXTRACT EPIAPC04 database
Successfully unregistered EXTRACT EPIAPC04 from database.

GGSCI (ggpqbopc04-a.sbg-qbo-ppd.a.intuit.com as ggs@PQB0PC04) 13> obey diroby/create_epiapc04_tmp.oby

GGSCI (ggpqbopc04-a.sbg-qbo-ppd.a.intuit.com as ggs@PQB0PC04) 14> dblogin USERIDALIAS ggsources

Successfully logged into database.

GGSCI (ggpqbopc04-a.sbg-qbo-ppd.a.intuit.com as ggs@PQB0PC04) 15> add extract epiapc04, INTEGRATED TRANLOG, SCN 5641774085

EXTRACT added.

GGSCI (ggpqbopc04-a.sbg-qbo-ppd.a.intuit.com as ggs@PQB0PC04) 16> add exttrail ./dirdat/piacpc04/tr extract epiapc04, MEGABYTES 500

EXTTRAIL added.

GGSCI (ggpqbopc04-a.sbg-qbo-ppd.a.intuit.com as ggs@PQB0PC04) 18> register EXTRACT EPIAPC04 database scn 5377816740

Extract EPIAPC04 successfully registered with database at SCN 5377816740.

Where 5377816740 is the last change# from the query

GGSCI (ggpqbopc04-a.sbg-qbo-ppd.a.intuit.com as ggs@PQB0PC04) 19> info all

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	STOPPED	EPIAPC04	00:00:00	00:03:36
EXTRACT	STOPPED	EPQBPC04	00:00:00	00:11:44

GGSCI (ggpqbopc04-a.sbg-qbo-ppd.a.intuit.com as ggs@PQB0PC04) 12> start mgr
Manager started.

GGSCI (ggpqbopc04-a.sbg-qbo-ppd.a.intuit.com as ggs@PQB0PC04) 21> start e*

Sending START request to MANAGER ...
EXTRACT EPIAPC04 starting

Sending START request to MANAGER ...
EXTRACT EPQBPC04 starting

GGSCI (ggpqbopc04-a.sbg-qbo-ppd.a.intuit.com as ggs@PQB0PC04) 22> info all

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	STARTING	EPIAPC04	00:00:00	00:05:26
EXTRACT	STARTING	EPQBPC04	00:00:00	00:13:33

GGSCI (ggpqbopc04-a.sbg-qbo-ppd.a.intuit.com as ggs@PQB0PC04) 23> !
info all

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	RUNNING	EPIAPC04	00:00:00	00:00:09
EXTRACT	RUNNING	EPQBPC04	00:00:00	00:00:10

- [Recreate and start all pumps on us-west-2](#)

```
GGSCI (ggpqbopc04-a.sbg-qbo-ppd.a.intuit.com) 6> obey diroby/create_ppqbosc04.oby

GGSCI (ggpqbopc04-a.sbg-qbo-ppd.a.intuit.com) 7> add extract ppqbosc04, EXTTRAILSOURCE ./dirdat/pqbopc04/tr
EXTRACT added.

GGSCI (ggpqbopc04-a.sbg-qbo-ppd.a.intuit.com) 8> add rmttrail ./dirdat/pqbopc04/tr, extract ppqbosc04, MEGABYTES
500
RMTTRAIL added.

GGSCI (ggpqbopc04-a.sbg-qbo-ppd.a.intuit.com as ggs@PQBOPC04) 33> obey diroby/create_ppiwpc04.oby

GGSCI (ggpqbopc04-a.sbg-qbo-ppd.a.intuit.com as ggs@PQBOPC04) 34> add extract ppiwpc04, EXTTRAILSOURCE ./dirdat
/piacpc04/tr
EXTRACT added.

GGSCI (ggpqbopc04-a.sbg-qbo-ppd.a.intuit.com as ggs@PQBOPC04) 35> add rmttrail ./dirdat/piacpc04/tr, extract
ppiwpc04, MEGABYTES 500
RMTTRAIL added.

GGSCI (ggpqbopc04-a.sbg-qbo-ppd.a.intuit.com as ggs@PQBOPC04) 36>

GGSCI (ggpqbopc04-a.sbg-qbo-ppd.a.intuit.com as ggs@PQBOPC04) 36> info all
```

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	RUNNING	EPIAPC04	00:00:01	00:00:04
EXTRACT	RUNNING	EPQBPC04	00:00:01	00:00:04
EXTRACT	STOPPED	PPIWPC04	00:00:00	00:00:08
EXTRACT	STOPPED	PPQBSC04	00:00:00	00:00:32

```
GGSCI (ggpqbopc04-a.sbg-qbo-ppd.a.intuit.com as ggs@PQBOPC04) 37> start p*

Sending START request to MANAGER ...
EXTRACT PPIWPC04 starting

Sending START request to MANAGER ...
EXTRACT PPQBSC04 starting
```

- [Alter the replicat on us-east-2](#)

```
GGSCI (ggpqbosc04-a.sbg-qbo-ppd.a.intuit.com) 2> info all
```

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
REPLICAT	STOPPED	RPQBPC04	00:00:00	00:19:49

```
GGSCI (ggpqbosc04-a.sbg-qbo-ppd.a.intuit.com) 3> info RPQBPC04
```

```
REPLICAT RPQBPC04 Last Started 2019-05-09 09:57 Status STOPPED
INTEGRATED
Checkpoint Lag      00:00:00 (updated 00:20:07 ago)
Log Read Checkpoint File /u01/ogg/pqbosc04/19.1.0.0.1/dirdat/pqbopc04/tr000027
                    2019-05-09 10:02:51.045581 RBA 122685
```

```
GGSCI (ggpqbosc04-a.sbg-qbo-ppd.a.intuit.com) 4> sh ls -trl /u01/ogg/pqbosc04/19.1.0.0.1/dirdat/pqbopc04/
```

```
total 4152
-rw-r----- 1 oracle oinstall 4248489 May  9 11:08 tr000000
```

```
GGSCI (ggpqbosc04-a.sbg-qbo-ppd.a.intuit.com) 5> alter replicat RPQBPC04 extseqno 0 extrba 0
```

```
2019-05-09 11:08:39 INFO    OGG-06594 Replicat RPQBPC04 has been altered. Even the start up position might be
updated, duplicate suppression remains active in next startup. To override duplicate suppression, start
RPQBPC04 with NOFILTERDUPTRANSACTIONS option.
```

```
REPLICAT (Integrated) altered.
```

- [Start the replicat from the Low Watermark CNS on us-east-2](#)

```
GGSCI (ggpqbosc04-a.sbg-qbo-ppd.a.intuit.com) 7> start replicat RPQBPC04 aftercsn 5377818197
```

```
Sending START request to MANAGER ...
REPLICAT RPQBPC04 starting
```

```
GGSCI (ggpqbosc04-a.sbg-qbo-ppd.a.intuit.com) 8> info RPQBPC04
```

REPLICAT	RPQBPC04	Last Started	2019-05-09 11:11	Status	RUNNING
INTEGRATED					
Checkpoint Lag	00:50:37 (updated 00:00:03 ago)				
Process ID	30816				
Log Read Checkpoint	File /u01/ogg/pqbosc04/19.1.0.0.1/dirdat/pqbopc04/tr000000				
	2019-05-09 10:20:59.038731 RBA 35036				

Note: 5377818197 is Low Watermark CSN value

4.3 Downgrade GG Hub on us-east-2

- [Stop the QBO pump on us-west-2](#)

```
GGSCI (ggpqbosc04-a.sbg-qbo-ppd.a.intuit.com as ggs@PQB0PC04) 24> stop PPQBSC04
```

```
Sending STOP request to EXTRACT PPQBSC04 ...
Request processed.
```

- [Wait until all trail files have been applied on us-west-2](#)

```
GGSCI (ggpqbosc04-a.sbg-qbo-ppd.a.intuit.com) 20> send RPQBPC04 status
```

```
Sending STATUS request to REPLICAT RPQBPC04 ...  
Current status: At EOF  
Sequence #: 0  
RBA: 4,293,393  
0 records in current transaction.
```

- Stop the replicat and get SCN on **us-east-2**

```
GGSCI (ggpqbosc04-a.sbg-qbo-ppd.a.intuit.com) 14> stop RPQBPC04
```

```
Sending STOP request to REPLICAT RPQBPC04 ...  
Request processed.
```

```
GGSCI (ggpqbosc04-a.sbg-qbo-ppd.a.intuit.com) 21> obey diroby/dblogin_t.oby
```

```
GGSCI (ggpqbosc04-a.sbg-qbo-ppd.a.intuit.com) 22> dblogin USERIDALIAS ggtarget
```

```
Successfully logged into database.
```

```
GGSCI (ggpqbosc04-a.sbg-qbo-ppd.a.intuit.com as ggt@PQBOSC04) 23> info RPQBPC04 detail
```

```
REPLICAT RPQBPC04 Last Started 2019-05-09 11:34 Status RUNNING  
INTEGRATED  
Checkpoint Lag 00:00:00 (updated 00:00:08 ago)  
Process ID 32112  
Log Read Checkpoint File /u01/ogg/pqbosc04/19.1.0.1/dirdat/pqbopc04/tr000000  
2019-05-09 11:33:33.039687 RBA 4293393
```

```
INTEGRATED Replicat  
DBLOGIN Provided, inbound server name is OGG$RPQBPC04 in ATTACHED state
```

```
Current Log BSN value: 5377818243
```

```
Low Watermark CSN value: 5377864559  
(All source transactions prior to this scn have been applied)
```

```
High Watermark CSN value: 5377864588  
(Some source transactions between this scn and the low watermark may have been applied)
```

Note: We will use Low Watermark CSN value (5377864559 in the example above) later on

- Clean up dirchk and dirdat under 12.3 on **us-east-2**

```
. /l/orcl  
[oracle@ggpqbosc04-a 12.1.2.1.6]$ rm -r /u01/ogg/${RDS_SID}/12.3.0.1.4/dirchk/*  
[oracle@ggpqbosc04-a 12.1.2.1.6]$ rm -r /u01/ogg/${RDS_SID}/12.3.0.1.4/dirdat/*
```

- Use the command in 4.1 above to switch from 19.1 to 12.3
- Use AWS Console to delete GG Hub Cloudformation stacks starting with goldengate-<RDS>-b and goldengate-<RDS>-b in **us-east-2**. Make sure you choose **us-east-2** region.
- Create new GG Hub with Goldengate 12.3 on **us-east-2**

```
chmod +x create_gg_hub_ha.sh
./create_gg_hub_ha.sh
Usage: ./create_gg_hub_ha.sh <AppName> <AppEnv> <DBName> <Region> <VPC> <HA> [<VolumeSize>]
    where <AppName> should be qbo, payments, etc
          <AppEnv> should be qa, e2e, stg, sbx, prf or prod
          <Region> should be us-west-2, us-east-2, etc
          <HA> should be yes or no
          <VPC> should be vpc-1, vpc-2, etc or Intuit-vpc-1, Intuit-vpc-2, etc
          [<VolumeSize>] is optional. It is EBS volume /u01 size in GB and default is 1000 GB
```

For non prod example

```
./create_gg_hub_ha.sh psp prf pspibob1 us-west-2 vpc-2 no
```

For prod, it would look like

```
./create_gg_hub_ha.sh psp prod pspspib1 us-east-2 vpc-1 no
```

For non-prod, the following command creates E2E C3 GG Hub in us-east-2

- Start manager on **us-east-2**
- Rollover and start the pump on **us-west-2**

```
GGSCI (ggpqbopc04-a.sbg-qbo-ppd.a.intuit.com) 5> alter extract PPQBSC04 etrollover
```

```
2019-05-09 13:13:00 INFO      OGG-01520 Rollover performed. For each affected output trail of Version 10 or
higher format, after starting the source extract, issue ALTER EXTSEQNO for that trail's reader (either pump
EXTRACT or REPLICAT) to move the reader's scan to the new trail file; it will not happen automatically.
EXTRACT altered.
```

```
GGSCI (ggpqbopc04-a.sbg-qbo-ppd.a.intuit.com) 2> start PPQBSC04
```

```
Sending START request to MANAGER ...
EXTRACT PPQBSC04 starting
```

- Create and start the replicat on **us-east-2**

```
GGSCI (ggpqbosc04-a.sbg-qbo-ppd.a.intuit.com) 2> obey diroby/dblogin_t.oby

GGSCI (ggpqbosc04-a.sbg-qbo-ppd.a.intuit.com) 3> dblogin USERIDALIAS ggtarget

Successfully logged into database.

GGSCI (ggpqbosc04-a.sbg-qbo-ppd.a.intuit.com as ggt@PQB0SC04) 5> obey diroby/create_rpqbpc04.oby

GGSCI (ggpqbosc04-a.sbg-qbo-ppd.a.intuit.com as ggt@PQB0SC04) 6> dblogin USERIDALIAS ggtarget

Successfully logged into database.

GGSCI (ggpqbosc04-a.sbg-qbo-ppd.a.intuit.com as ggt@PQB0SC04) 7> add replicat rpqbpc04, INTEGRATED, EXTTRAIL .
/dirdat/pqbopc04/tr

REPLICAT (Integrated) added.

GGSCI (ggpqbosc04-a.sbg-qbo-ppd.a.intuit.com as ggt@PQB0SC04) 8> sh ls dirdat/pqbopc04/

tr000001
GGSCI (ggpqbosc04-a.sbg-qbo-ppd.a.intuit.com as ggt@PQB0SC04) 9> alter replicat RPQBPC04 extseqno 1 extrba 0
REPLICAT (Integrated) altered.

GGSCI (ggpqbosc04-a.sbg-qbo-ppd.a.intuit.com as ggt@PQB0SC04) 11> start replicat RPQBPC04 aftercsn 5377864559

Sending START request to MANAGER ...
REPLICAT RPQBPC04 starting
```

Note: 5377864559 is Low Watermark CSN value.